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Unit – 4 : Session – 5 : SLO – 1 | Implementation of Tree Using Array — Concepts

■ I. Fill in the Blanks

- The array representation of binary tree can be done using a technique called SEQUENTIAL (LEVEL-ORDER) REPRESENTATION.
- In level-order traversal, the elements of the binary tree are stored in the array in the order in which they are visited in a BREADTH-FIRST (LEVEL-ORDER) TRAVERSAL.
- The array representation of binary tree allows for efficient RANDOM ACCESS to the elements of the tree.
- If a binary tree has n nodes, the array representation of the tree can be stored in an array of size n , allowing for DIRECT ($O(1)$) access to each node in the tree.
- The ROOT of the binary tree is stored at the first position of the array.
- The left child of node i can be found at position $2*i$.
- The right child of node i can be found at position $2*i + 1$.
- The parent of node i can be found at position $\text{floor}(i/2)$.
- The array representation of binary tree can be useful in algorithms that require fast access to the elements of a tree, such as HEAP SORT or PRIORITY QUEUE (searching) algorithms.
- It is important to note that not all binary trees can be EFFICIENTLY represented in an array, as the SIZE (number of nodes) in the array must be known in advance.